

Binary64 construction of tanh approximants by scaled truncated SVD

Working proof for the numerical-recovery theorem

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Scope. This is a new numerical-recovery result taking the revised boundary-corrected representation theorem as its approximation input. It does not reprove that representation theorem, and it is not a theorem for every unscaled default `1stsq` call. The algorithm specified here uses binary64 for matrix assembly, SVD, coefficient formation, evaluation, and certification. It uses no higher-precision solve, exact accumulator, double-double representation, or exact-rational optimization oracle. Exact orthogonal matrices introduced in the proof are analytical comparison objects, not computed intermediates. The result is a computable upper error bound, not a lower bound on the best possible numerical error.

1 Approximation input and exactly stored hidden geometry

Let $\Omega = [-1, 1]$, $I_\Delta = [-1 - \Delta, 1 + \Delta]$, and suppose f is real on I_Δ and has a holomorphic extension bounded by B on $U_\delta(I_\Delta) = \{z : \text{dist}(z, I_\Delta) < \delta\}$. Take $h = 2/N$, $x_j = -1 + jh$, $j = -R, \dots, N + R$, $\gamma = \lambda/h$, and $W = N + 2R + 1$. There are $n = W + 1$ readout coordinates, including the output bias. Write $\varphi_b = 1$, $\varphi_j(x) = \tanh(\gamma x + \beta_j)$, $\beta_j = -\gamma x_j$, and $q_c = \sum_j c_j \varphi_j$.

Assume the representation theorem supplies positive, explicitly computable coordinate envelopes and an approximation bound

$$|c_{*,j}| \leq B\alpha_j, \quad \|f - q_{c_*}\|_{L^\infty(\Omega)} \leq E_W. \quad (1)$$

E_W is a known bound, not the unknown actual error. It may be an outward-rounded upper bound on the three-term formula or on the sharper uncompressed formula. Its numerical evaluation error is retained in the final budget.

For example, the envelopes follow from the reference construction in the project note [1]. Put

$$d_{\text{pole}} = \frac{\pi h}{2\lambda}, \quad m = \left\lfloor \frac{R+1}{2} \right\rfloor, \quad D_\lambda = \frac{\pi}{2\lambda} + \frac{4 \log 2}{\pi}.$$

With $\zeta = e^{-2\lambda}$, $p_0 = 1$, $p_k = \prod_{\ell=1}^k (1 - \zeta^\ell)$, define

$$L_{i,m} = \frac{\zeta^{[i(i+1)-1]/2}}{p_{i-1}p_{m-i}} \prod_{\substack{1 \leq j \leq m \\ j \neq i}} (1 + \zeta^{j-1/2}), \quad C_m = \sum_{i=1}^m L_{i,m}.$$

Start each output-weight envelope at $h/[2(\delta - d_{\text{pole}})]$. At corrected slot i on either side, add $hD_\lambda L_{i,m}/(2\delta)$. Set $\alpha_b = 1 + \sum_{\text{weight slots}} \alpha_j$. Then

$$K_W^{\text{ref}} := \sum_j \alpha_j = 1 + \frac{Wh}{\delta - d_{\text{pole}}} + \frac{2hD_\lambda C_m}{\delta}. \quad (2)$$

Indeed, the baseline density is bounded by $B/(\delta - d_{\text{pole}})$; each boundary moment is bounded by BhD_λ/δ ; $L_{i,m}$ is the absolute row sum of its coefficient map; the conversion to a tanh weight has a factor $1/2$; and the anchored bias is bounded by $B + \sum_j |w_j|$. This proves the coordinatewise assertion from the coefficient formulas of the representation theorem without evaluating those functionals numerically.

For the first binary64 specialization, take

$$\lambda = 1/4, \quad N = 2^k, \quad s = 2^\ell, \quad \gamma = N/8, \quad \beta_j = N/8 - j/4, \quad z_i = -1 + 2i/(Ns), \quad i = 0, \dots, Ns. \quad (3)$$

Assume the dyadic quantities in (3) are exactly representable, $M := Ns + 1 \geq n$, and all integer indices are in their exact range. These are finite range checks. The analytic admissibility conditions remain those of the representation theorem. More generally, the proof applies to any exactly stored hidden geometry satisfying the same approximation input and the sampling conditions below.

Define

$$\|g\|_X^2 = \frac{1}{M} \sum_{i=0}^{M-1} |g(z_i)|^2, \quad \|g\|_{2,\Omega}^2 = \frac{1}{2} \int_{-1}^1 |g(x)|^2 dx.$$

The algorithm receives binary64 labels y_i^{fl} satisfying $|y_i^{\text{fl}} - f(z_i)| \leq \eta_{\text{lab}}$. This is the error bound for the specified real-valued target-evaluation routine or supplied observations. It is not access to derivatives or to complex target values.

2 Binary64 arithmetic and enclosures

All construction and evaluation arithmetic is binary64, round-to-nearest, with $u = 2^{-53}$, gradual underflow, and no overflow or invalid operations. Use $\mu = 2^{-1074}$, the smallest positive binary64 number, so

$$|\text{RN}(t) - t| \leq u|t| + \mu. \quad (4)$$

Using μ rather than half that quantity makes the absolute allowance itself representable. Fused multiply-add performs one rounding. Certification uses outward-rounded binary64 interval endpoints; all bounds below are rounded in a direction that preserves the indicated inequality. In particular, a denominator lower bound is rounded downward and a numerator upper bound upward. No interval endpoint has more than 53 significand bits.

2.1 Pairwise dot products

Let pdot form rounded products and add them in a balanced binary tree. For $k \geq 1$, set

$$\ell_k = \lceil \log_2 k \rceil, \quad g_t = \frac{tu}{1 - tu}, \quad a_k = g_{\ell_k+1}, \quad b_k = \frac{(2k-1)\mu}{1 - (\ell_k+1)u}.$$

All denominators are assumed positive. Empty dot products are zero.

Lemma 1 (Pairwise dot-product error). *For binary64 vectors v, w of length k ,*

$$|\text{pdot}(v, w) - v^T w| \leq a_k \sum_{i=1}^k |v_i w_i| + b_k.$$

Proof. Write the error of each operation as a relative factor of modulus at most u and an additive error of modulus at most μ , using (4). Each exact product passes through at most $\ell_k + 1$ rounded operations, so its multiplicative error is bounded by g_{ℓ_k+1} . There are at most $2k - 1$ additive errors. Each is amplified by at most $(1 + u)^{\ell_k+1} \leq [1 - (\ell_k + 1)u]^{-1}$. Expanding the tree and applying the triangle inequality gives the assertion. $\square$

2.2 Activation routine

Let t_{64} be a fixed binary64 tanh routine with a proved absolute-error bound

$$|t_{64}(z) - \tanh z| \leq \varepsilon_t \quad (z \in \mathbb{F}_{64} \text{ finite}). \quad (5)$$

Clamp its result to $[-1, 1]$. Section 6 supplies a fully binary64 example with $\varepsilon_t = 1024u$; a more accurate validated routine may be substituted without changing the proof.

Lemma 2 (Fused-argument activation error). *If $\widehat{\varphi}_j(x) = \text{clamp}_{[-1,1]} t_{64}(\text{RN}(\gamma x + \beta_j))$ uses a fused affine argument at a binary64 input, then*

$$|\widehat{\varphi}_j(x) - \varphi_j(x)| \leq \varepsilon_\varphi := \varepsilon_t + \frac{u}{2(1-u)} + \mu.$$

Proof. Write the rounded affine argument as $z(1 + \theta) + e$, with $|\theta| \leq u$, $|e| \leq \mu$. The additive perturbation changes $\tanh$ by at most μ . On the multiplicative segment,

$$|\tanh(z(1 + \theta)) - \tanh z| \leq u|z| \text{sech}^2((1 - u)|z|) \leq \frac{u}{2(1 - u)},$$

because $t \text{sech}^2 t \leq 1/2$ for $t \geq 0$. Add (5). Clamping cannot increase the error relative to a number in $[-1, 1]$. $\square$

2.3 Computable interval bounds

Point intervals for stored numbers, followed by outward-rounded operations, enclose the exact expressions in those numbers. This follows inductively from monotonic enclosure for each arithmetic operation; interval multiplication uses the minimum and maximum of the four endpoint products. Square roots use their monotonicity. Thus an interval matrix $\mathcal{H}$ enclosing H gives the bound

$$\|H\|_2 \leq \|H\|_F \leq \left(\sum_{ij} \max\{|\mathcal{H}_{ij}|, |\overline{\mathcal{H}}_{ij}|\}^2 \right)^{1/2},$$

evaluated upward. Balanced interval summation or Lemma 1 can be used to reduce overestimation. The certificate does not require a spectral-norm oracle.

The positive coordinate envelopes and E_W can likewise be enclosed in binary64 using finite products, rational operations, and interval elementary functions. Exponentials can be enclosed by scaling their argument to at most $1/2$, evaluating a Taylor polynomial with its explicit remainder, and repeated outward squaring. Very small positive values are rounded upward rather than silently set to zero. For instance, constants can be enclosed using

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239), \quad \log 2 = 2 \sum_{j=0}^J \frac{3^{-(2j+1)}}{2j+1} + R_J, \quad 0 < R_J \leq \frac{9 \cdot 3^{-(2J+3)}}{4(2J+3)}.$$

Use the alternating arctangent series with its next-term remainder; the first identity follows from the tangent addition formula, and the second follows by integrating the geometric series for $2/(1-t^2)$ from 0 to $1/3$. Finite outward-rounded sums enclose these expressions. An enclosure is not required to shrink below binary64 resolution. If a denominator enclosure is nonpositive, or an enclosure overflows, the chosen design is not certified by that computation.

3 The binary64 recovery algorithm

3.1 Scaling and matrix assembly

Choose positive binary64 d_j with $d_j^2 \geq \alpha_j$, by outward square roots. Let

$$D = \text{diag}(d_j), \quad L \geq \left(\sum_j d_j^2 \right)^{1/2}, \quad S \geq BL,$$

where L, S are upward-rounded binary64 bounds. The mathematical reference matrices, never evaluated exactly by the algorithm, are

$$A_{ij} = \varphi_j(z_i), \quad F = AD/\sqrt{M}, \quad y_X = (f(z_i))_i/\sqrt{M}.$$

Compute $r_0 = \text{RN}(1/\text{RN}(\sqrt{M}))$ and

$$\bar{F}_{ij} = \text{RN}(r_0 \text{RN}(d_j \hat{\varphi}_j(z_i))), \quad \bar{y}_i = \text{RN}(r_0 y_i^{\text{fl}}).$$

A sufficient pair of assembly bounds is

$$A_0 = (\varepsilon_\varphi + g_4)L + 3\mu\sqrt{Mn}, \quad (6)$$

$$B_0 = \eta_{\text{lab}} + g_3(B + \eta_{\text{lab}}) + \mu\sqrt{M}, \quad (7)$$

so that $\|\bar{F} - F\|_2 \leq A_0$ and $\|\bar{y} - y_X\|_2 \leq B_0$. Tighter interval enclosures may replace these bounds. The letter B_0 denotes data error, not the analytic magnitude B .

3.2 Factorization and its checks

Compute a thin binary64 SVD, for example `DGESVD`, obtaining $\hat{U} \in \mathbb{R}^{M \times n}$, $\hat{V} \in \mathbb{R}^{n \times n}$, and a nonnegative diagonal $\hat{\Sigma}$; let $\hat{\sigma}_1$ be its largest diagonal entry. Compute outward bounds

$$\omega_U \geq \|\hat{U}^T \hat{U} - I\|_2, \quad \omega_V \geq \|\hat{V}^T \hat{V} - I\|_2, \quad (8)$$

$$r_F \geq \|\bar{F} - \hat{U} \hat{\Sigma} \hat{V}^T\|_2. \quad (9)$$

Frobenius-norm bounds from interval matrix products suffice. Require $\omega_U, \omega_V \leq 1/2$. Define

$$d_U = \frac{\omega_U}{1 + \sqrt{1 - \omega_U}}, \quad d_V = \frac{\omega_V}{1 + \sqrt{1 - \omega_V}}, \quad v_+ = \sqrt{1 + \omega_V}, \quad u_+ = \sqrt{1 + \omega_U},$$

and

$$a = A_0 + r_F + \hat{\sigma}_1(d_U v_+ + d_V). \quad (10)$$

No unspecified backward-error constant for the SVD occurs in this definition.

3.3 Projection, cutoff, and coefficient export

Compute all n projected coordinates

$$\hat{p}_i = \text{pdot}(\hat{U}_{:,i}, \bar{y}).$$

Take upward bounds $Y \geq \|\bar{y}\|_2$ and $P \geq \|\hat{p}\|_2$. Define

$$e_p^{\text{fp}} = a_M u_+ \sqrt{n} Y + b_M \sqrt{n}, \quad (11)$$

$$e_d^{\text{fp}} = uP + \mu \sqrt{n} \hat{\sigma}_1, \quad (12)$$

$$b = B_0 + d_U Y + e_p^{\text{fp}} + e_d^{\text{fp}}. \quad (13)$$

The projection bound can alternatively use a tighter interval evaluation of $\hat{p} - \hat{U}^T \bar{y}$.

Choose a positive binary64 regularization allowance ε_{reg} , and choose a binary64 cutoff $\tau > 0$ by outward arithmetic so that

$$\tau S \geq E_W + aS + b + \varepsilon_{\text{reg}}. \quad (14)$$

Set $J = \{i : \hat{\sigma}_i > \tau\}$ and compute

$$\hat{z}_i = \text{RN}(\hat{p}_i / \hat{\sigma}_i) \quad (i \in J), \quad \bar{c}_j = \text{RN}\left(d_j \text{pdot}(\hat{V}_{j,J}, \hat{z}_J)\right).$$

An empty retained set gives the zero vector. Choose an upward bound

$$\rho \geq \tau S - (E_W + aS + b). \quad (15)$$

This is the actual cutoff allowance, including cutoff-construction rounding.

For the export bound, set

$$D_1 \geq \sum_j d_j, \quad H \geq \left\| \left| \hat{V} \right|^T d \right\|_2, \quad e_n = u + (1 + u)a_n,$$

where $d = (d_j)_j$ and matrix absolute values are entrywise. Define

$$e_c = 2LSd_V + 2Se_n H + (1 + u)b_n D_1 + n\mu, \quad T = 2LS + e_c, \quad R_X = 2E_W + \rho + 4aS + 3b + e_c. \quad (16)$$

All these quantities are computed with binary64 upper enclosures. In exact scaling arithmetic, $LS = BK_W$; actual stored scales and their rounded norm bounds are retained here instead.

Failure handling. The high-accuracy branch requires finite computed quantities, no overflow, a successful finite factorization, and the displayed orthogonality checks. Failure is explicitly reported, not repaired by increasing precision. A total implementation may return the zero network and the valid error bound B in such a case. The geometric corollary below concerns certified SVD runs, not this coarse fallback. LAPACK's explicit failure status is part of its interface [2]. No claim of success for every library call is needed for the validity of the bounds on certified runs.

4 The binary64 recovery theorem

For $\varrho > 1$ and integer $p \geq 1$, assume

$$\lambda(\varrho - \varrho^{-1}) \leq \pi, \quad \frac{h}{4}(\varrho + \varrho^{-1}) < \delta, \quad s \geq 2(p + 1)^2. \quad (17)$$

At an arbitrary real input, let $Q(x)$ be its nearest binary64 value clamped to $[-1, 1]$. Take $\xi \geq \sup_{x \in \Omega} |Q(x) - x|$; for example $\xi = u + \mu$ is sufficient. Let

$$\chi = \frac{2p^2(p+1)\xi}{h}, \quad C_X = (1 + \chi) \sqrt{\frac{2M}{M-1}}, \quad C_A = \frac{2[1 + \sqrt{2}(1 + \chi)]}{\varrho - 1} \varrho^{-p}. \quad (18)$$

Evaluate the exported network at $Q(x)$ using fused affine arguments, t_{64} , clamping the activations, rounded products, and pairwise summation including the bias. Denote the computed result by $\hat{q}(x)$.

Theorem 3 (Scaled truncated-SVD construction entirely in binary64). *Under the approximation input (1), the geometric and numerical conditions above, and the specified binary64 algorithm,*

$$\|\bar{c}\|_1 \leq T, \quad (19)$$

$$\|f - \hat{q}\|_{2,\Omega} \leq C_X R_X + C_A(B + T) + \frac{B\xi}{\delta} + (\varepsilon_\varphi + a_n)T + b_n. \quad (20)$$

In particular, the coefficient cap and sampled residual are conclusions of the SVD construction, not conditions imposed on a returned LS candidate. Every error quantity in (20) has a binary64 enclosure defined above. No lower bound on the least singular value of the full feature matrix or spectral gap at the cutoff is required.

5 Proof of Theorem 3

5.1 The reference vector in scaled coordinates

Let $a_* = D^{-1}c_*$. Since $d_j^2 \geq \alpha_j$,

$$\|a_*\|_2^2 \leq B^2 \sum_j \frac{\alpha_j^2}{d_j^2} \leq B^2 \sum_j d_j^2 \leq B^2 L^2 \leq S^2.$$

Also $\|Fa_* - y_X\|_2 \leq E_W$. For every a ,

$$\|Da\|_1 \leq \|d\|_2 \|a\|_2 \leq L \|a\|_2. \quad (21)$$

The normalized feature matrix satisfies $\|F\|_2 \leq \|F\|_F \leq L$. These statements use the envelopes but never evaluate the reference coefficients.

5.2 Assembly errors

Writing the two normalization roundings as multiplicative factors gives $r_0 = M^{-1/2}(1 + \theta_1)/(1 + \theta_0)$, $|\theta_i| \leq u$. Combining them with the two product roundings yields a relative factor bounded by g_4 . Lemma 2 and activation clamping imply, entrywise,

$$|\bar{F}_{ij} - F_{ij}| \leq \frac{d_j}{\sqrt{M}}(\varepsilon_\varphi + g_4) + 3\mu.$$

The 3μ accounts for both possibly subnormal product roundings; the normalization is normal under the finite index-range assumptions. Taking the Frobenius norm proves (6). Similarly,

$$|\bar{y}_i - f(z_i)/\sqrt{M}| \leq \frac{\eta_{\text{lab}} + g_3(B + \eta_{\text{lab}})}{\sqrt{M}} + \mu,$$

which proves (7).

5.3 An exact nearby factorization for analysis only

Define the polar factors

$$U_0 = \widehat{U}(\widehat{U}^T \widehat{U})^{-1/2}, \quad V_0 = \widehat{V}(\widehat{V}^T \widehat{V})^{-1/2}.$$

The checks (8) make the two Gram matrices positive definite. Thus $U_0^T U_0 = I$ and $V_0^T V_0 = I$. The singular values of $\widehat{U}$ lie in $[\sqrt{1 - \omega_U}, \sqrt{1 + \omega_U}]$, giving

$$\|\widehat{U} - U_0\|_2 \leq d_U, \quad \|\widehat{V} - V_0\|_2 \leq d_V, \quad \|\widehat{V}\|_2 \leq v_+.$$

Let $F_0 = U_0 \widehat{\Sigma} V_0^T$. Then

$$\begin{aligned} \|F - F_0\|_2 &\leq A_0 + r_F + \left\| (\widehat{U} - U_0) \widehat{\Sigma} \widehat{V}^T \right\|_2 + \left\| U_0 \widehat{\Sigma} (\widehat{V} - V_0)^T \right\|_2 \\ &\leq A_0 + r_F + \widehat{\sigma}_1(d_U v_+ + d_V) = a. \end{aligned}$$

U_0, V_0 , and F_0 are never numerically computed. They show that the returned binary64 factors describe a nearby exactly orthogonal singular decomposition. Unlike a singular-vector perturbation theorem, this argument needs no gap in the original spectrum.

5.4 Floating-point projection and division become a bounded data perturbation

Lemma 1 gives

$$\|\widehat{p} - \widehat{U}^T \overline{y}\|_2 \leq a_M \left\| \left| \widehat{U} \right|^T \left| \overline{y} \right| \right\|_2 + b_M \sqrt{n} \leq a_M u_+ \sqrt{n} Y + b_M \sqrt{n} = e_p^{\text{fp}}.$$

For retained indices, division rounding implies

$$|\widehat{\sigma}_i \widehat{z}_i - \widehat{p}_i| \leq u |\widehat{p}_i| + \mu \widehat{\sigma}_i.$$

Consequently the Euclidean norm of these discrepancies is at most e_d^{fp} . Let

$$w = \widehat{\Sigma}_J \widehat{z}_J - U_{0,J}^T \overline{y}, \quad y_0 = \overline{y} + U_{0,J} w.$$

Then

$$\|y_0 - y_X\|_2 \leq B_0 + d_U Y + e_p^{\text{fp}} + e_d^{\text{fp}} = b, \quad U_{0,J}^T y_0 = \widehat{\Sigma}_J \widehat{z}_J.$$

Thus $a_0 = V_{0,J} \widehat{z}_J$ is exactly the truncated-SVD solution of $F_0 a \approx y_0$, with cutoff τ . This remains true for an empty retained set. Again y_0 is an analytical comparison vector, not a high-precision input.

5.5 The SVD comparison and adaptive cutoff

Put $\eta = E_W + aS + b$. Then

$$\|F_0 a_* - y_0\|_2 \leq \eta.$$

Write $y_0 = F_0 a_* + e$, $\|e\|_2 \leq \eta$. For exact orthogonal singular factors,

$$a_0 = V_{0,J} V_{0,J}^T a_* + (F_0)_\tau^\dagger e.$$

Hence

$$\|a_0\|_2 \leq S + \eta/\tau \leq 2S, \quad \|\widehat{z}_J\|_2 = \|a_0\|_2 \leq 2S.$$

The discarded singular values of F_0 are at most τ , so

$$\|F_0 a_0 - y_0\|_2 \leq \eta + \tau S.$$

Returning to the original sample problem gives

$$\begin{aligned} \|F a_0 - y_X\|_2 &\leq \eta + \tau S + a \|a_0\|_2 + b \\ &\leq E_W + \tau S + 3aS + 2b \\ &\leq 2E_W + 4aS + 3b + \rho. \end{aligned}$$

The numerical choice (14), rather than an exact equality for an unrounded cutoff, is what is used in these inequalities.

5.6 Export into native tanh coefficients

First,

$$\left\| D(\widehat{V}_J - V_{0,J}) \widehat{z}_J \right\|_1 \leq L d_V \|\widehat{z}_J\|_2 \leq 2LS d_V.$$

For the computed row dot product and multiplication by d_j , the rounding model and Lemma 1 give

$$|\bar{c}_j - d_j(\widehat{V}_J \widehat{z}_J)_j| \leq e_n d_j \sum_{i \in J} |\widehat{V}_{ji} \widehat{z}_i| + (1+u)b_n d_j + \mu.$$

Sum over j and use

$$\sum_j d_j \sum_{i \in J} |\widehat{V}_{ji} \widehat{z}_i| \leq H \|\widehat{z}_J\|_2 \leq 2SH.$$

It follows that

$$\|\bar{c} - D a_0\|_1 \leq e_c. \quad (22)$$

Equation (21) now gives $\|\bar{c}\|_1 \leq 2LS + e_c = T$. Since $|\varphi_j(x)| \leq 1$ on the real axis, export changes either sampled RMS error or uniform error by at most e_c . Therefore

$$\|f - q\bar{c}\|_X \leq R_X. \quad (23)$$

This proves the coefficient conclusion and the sampled-residual conclusion without assuming them of the returned readout.

5.7 Direct continuous- L^2 sampling lemma

Let $g = f - q\bar{c}$. On every core cell, the closed Bernstein ellipse of parameter ϱ lies in the target domain and in the strip where each tanh feature has magnitude at most one. Indeed,

$$|\tanh(u + iv)|^2 = \frac{\sinh^2 u + \sin^2 v}{\sinh^2 u + \cos^2 v} \leq 1 \quad (|v| \leq \pi/4),$$

and (17) gives the required scaled strip height. Thus $|g| \leq B + T$ on each ellipse. Expanding after the Joukowski map gives Chebyshev coefficients of magnitude at most $2(B+T)\varrho^{-k}$. Truncation at degree p gives a real polynomial P_I on each cell I such that

$$\|g - P_I\|_{\infty, I} \leq e_p, \quad e_p = \frac{2(B+T)}{\varrho - 1} \varrho^{-p}. \quad (24)$$

This follows by summing the geometric tail; it is also the local approximation step used in the project sampling proof [1].

For a polynomial P of degree at most p on a cell of length h , use the normalized cell norm $\|P\|_{2,I}^2 = h^{-1} \int_I |P|^2$. In the orthonormal Legendre basis $\sqrt{2j+1} P_j$ on $[-1, 1]$ with measure $dx/2$, the squared derivative norms sum to

$$\sum_{j=1}^p \frac{(2j+1)j(j+1)}{2} = \frac{p(p+1)^2(p+2)}{4}.$$

To see the derivative identity, use $P'_j = \sum_{k < j, j-k \text{ odd}} (2k+1) P_k$ and Legendre orthogonality. Bounding the differentiation operator by its Frobenius norm and rescaling gives

$$\|P'\|_{2,I} \leq \frac{(p+1)\sqrt{p(p+2)}}{h} \|P\|_{2,I}. \quad (25)$$

These polynomial identities can also be found from the classical orthogonal polynomial formulas in [4].

Let $\|P\|_{\text{trap},I}^2$ denote the composite trapezoidal approximation to $h^{-1} \int_I |P|^2$ using the $s+1$ cell observation points. On a subinterval of length h/s , integration by parts bounds the trapezoidal remainder for a differentiable w by $(h/(2s)) \int |w'|$. With $w = |P|^2$, summing and applying Cauchy–Schwarz gives

$$\left| \|P\|_{\text{trap},I}^2 - \|P\|_{2,I}^2 \right| \leq \frac{h}{s} \|P\|_{2,I} \|P'\|_{2,I} \leq \frac{(p+1)\sqrt{p(p+2)}}{s} \|P\|_{2,I}^2.$$

The last coefficient is at most $1/2$ by (17). Therefore

$$\|P\|_{2,I} \leq \sqrt{2} \|P\|_{\text{trap},I}. \quad (26)$$

Without input rounding, (24) and (26) imply $\|g\|_{2,I} \leq \sqrt{2} \|g\|_{\text{trap},I} + (1 + \sqrt{2})e_p$. Minkowski's inequality combines the cells. Globally,

$$\|g\|_{\text{trap}}^2 = \frac{1}{M-1} \left(\frac{|g(z_0)|^2 + |g(z_{M-1})|^2}{2} + \sum_{i=1}^{M-2} |g(z_i)|^2 \right) \leq \frac{M}{M-1} \|g\|_X^2.$$

The fit itself remains equally weighted LS: trapezoidal weights are only a proof device.

5.8 Input rounding and binary64 network evaluation

Every core-cell endpoint is exactly representable, so monotonicity of round-to-nearest implies $Q(I) \subseteq I$. For a degree- p polynomial, the Legendre expansion gives $\|P\|_{\infty,I} \leq (p+1) \|P\|_{2,I}$. The same derivative identity used above gives $\left\| \frac{P'_j}{j} \right\|_{\infty} \leq j(j+1)/2$ on $[-1, 1]$. Cauchy–Schwarz and rescaling therefore give $\|P'\|_{\infty,I} \leq p(p+1)^2 \|P\|_{2,I}/h \leq 2p^2(p+1) \|P\|_{2,I}/h$. Thus

$$\|P\|_{\infty,I} \leq (p+1) \|P\|_{2,I}, \quad \|P'\|_{\infty,I} \leq \frac{2p^2(p+1)}{h} \|P\|_{2,I}.$$

Hence $\|P \circ Q - P\|_{2,I} \leq \chi \|P\|_{2,I}$ and $\|P \circ Q\|_{2,I} \leq (1 + \chi) \|P\|_{2,I}$. Use (24) and (26) again:

$$\|g \circ Q\|_{2,I} \leq \sqrt{2}(1 + \chi) \|g\|_{\text{trap},I} + [1 + \sqrt{2}(1 + \chi)]e_p.$$

Combining cells gives

$$\|g \circ Q\|_{2,\Omega} \leq C_X \|g\|_X + C_A(B + T). \quad (27)$$

Cauchy's estimate on the target gives $\|f'\|_{\infty,\Omega} \leq B/\delta$, so $\|f - f \circ Q\|_{2,\Omega} \leq B\xi/\delta$.

At a representable input, activation approximation contributes at most $\varepsilon_\varphi \sum_{j \neq b} |\bar{c}_j|$. Applying Lemma 1 to the final dot product, including the constant feature, gives the additional bound $a_n T + b_n$. Thus

$$|\widehat{q}(x) - q_{\bar{c}}(Q(x))| \leq (\varepsilon_\varphi + a_n)T + b_n.$$

Finally write

$$f - \widehat{q} = (f - f \circ Q) + (g \circ Q) + (q_{\bar{c}} \circ Q - \widehat{q}),$$

and use (23) and (27). This proves (20) and completes the proof. $\square$

6 A fully binary64 activation kernel

The theorem permits a validated faster kernel. The following elementary example shows that its activation contract can be met without increasing working precision; the constant is deliberately conservative.

Lemma 4 (A binary64-only tanh routine). *There is a finite routine using binary64 addition, multiplication, division, fused multiply-add, comparisons, and exact power-of-two scaling such that*

$$|t_{64}(z) - \tanh z| \leq 1024u$$

for every finite binary64 z .

Proof. For $|z| \leq 2^{-20}$ return z ; for $|z| \geq 20$ return $\text{sign}(z)$. The first error is at most $|z|^3/3 < u$, since $|\tanh t| \leq |t|$ and $1 - \tanh'(t) = \tanh^2 t$. The second is at most $2e^{-40} < u$.

Otherwise set $t = 2|z|$ exactly. Choose the smallest integer $k \geq 0$ for which $r = t/2^k \leq 1/2$; here $k \leq 7$. Generate the coefficients $c_0 = 1$ and $c_j = \text{RN}(c_{j-1}/j)$ for $j = 1, \dots, 20$. Evaluate $\sum_{j=0}^{20} c_j(-r)^j$ by fused Horner evaluation and clamp the result to $[0, 1]$, obtaining q_0 . Square k times in binary64, obtaining q . Return the signed and clamped binary64 evaluation of $(1 - q)/(1 + q)$.

Coefficient formation has relative error at most g_{20} , and the Horner forward error is bounded by its absolute coefficient sum times g_{20} . Together they give an absolute polynomial-evaluation error at most $g_{40}e^r$. The alternating Taylor remainder is at most $r^{21}/21! < u$. Since $r \leq 1/2$, $e^r < 5/3$, $e^{-r} > 3/5$, and $g_{40} < 41u$,

$$q_0 = e^{-r}(1 + \delta_0), \quad |\delta_0| \leq 128u.$$

Clamping cannot enlarge the error. All nonzero kernel intermediates in this branch are normal and safely in range.

The squared result satisfies

$$q = e^{-t}(1 + \delta_0)^{2^k} \prod_{j=1}^k (1 + \delta_j)^{2^{k-j}}, \quad |\delta_j| \leq u.$$

Therefore $|\log(q/e^{-t})| \leq 130u2^k < 1/8$. Since $|e^a - 1| \leq |a|/(1 - |a|)$ for $|a| < 1$,

$$|q - e^{-t}| \leq \frac{8}{7} 130u 2^k e^{-t} < 256u.$$

For $k = 0$, $2^k e^{-t} \leq 1$; for $k \geq 1$, minimality of k gives $2^k < 4t$, so $2^k e^{-t} < 4/e < 3/2$. On $[0, 1]$, the derivative of $(1 - q)/(1 + q)$ has magnitude at most 2. The last subtraction, addition, and division add at most $g_3 < 4u$ in absolute error, because the exact ratio has magnitude at most one. Thus the total error is below $512u + 4u < 1024u$. Sign changes and final clamping do not increase it. $\square$

Remark 5. This kernel is an existence example for the all-binary64 arithmetic model, not a claim that its worst-case constant is close to a practical library's error. Using a sharper validated kernel reduces ε_t throughout the construction and evaluation bounds. No exact accumulation has been used: all dot products use the pairwise bound with their explicit length dependence.

7 Geometric convergence above a computed precision budget

Corollary 6 (Computed geometric-to-floor bound). *Consider a family of certified runs with the bandwidth and halo schedule of the representation corollary. Suppose the outward approximation bounds satisfy*

$$E_W \leq A_{\text{rep}} B e^{-c_{\text{app}} W} + \varepsilon_{\text{alias}, W} + \varepsilon_{E, W},$$

where $\varepsilon_{E, W}$ includes numerical overestimation in forming E_W . Let $C_X \leq \overline{C}_X$ on the stated finite binary64 range. Define

$$\begin{aligned} \mathcal{F}_W = C_X [2\varepsilon_{\text{alias}, W} + 2\varepsilon_{E, W} + \rho + 4aS + 3b + e_c] + C_A(B + T) + B\xi/\delta \\ + (\varepsilon_\varphi + a_n)T + b_n. \end{aligned}$$

If the computed bounds give $\mathcal{F}_W \leq \varepsilon_{\text{floor}}$ throughout that range, then

$$\|f - \hat{q}_W\|_{2, \Omega} \leq 2\overline{C}_X A_{\text{rep}} B e^{-c_{\text{app}} W} + \varepsilon_{\text{floor}}.$$

For $\varepsilon > \varepsilon_{\text{floor}}$, a sufficient admissible width is

$$W \geq \frac{1}{c_{\text{app}}} \log \frac{2\overline{C}_X A_{\text{rep}} B}{\varepsilon - \varepsilon_{\text{floor}}},$$

provided that width lies within the certified range.

Proof. Substitute the representation bound into Theorem 3, isolate its geometric term, and bound the displayed remainder. Solving the resulting inequality gives the width statement. $\square$

What is and is not asserted. The factorization defects, projection rounding, coefficient export, and inference errors are computed in the same fixed format as the construction. They need not be arbitrarily small, and their width dependence has not been hidden. The floor is an upper attainable-error budget for this implementation, not a necessary lower numerical limit. The theorem does not establish an unconditional 10^{-14} tolerance for every target or every binary64 library call. It provides a complete formula for deciding whether a specified binary64 run certifies that tolerance. No higher-precision escape is part of the stated algorithm.

References

- [1] Project proof note, *fixed_lambda_tanh_three_theorems.tex*. The revised boundary-corrected reference construction, local density bound, finite boundary-coefficient map, and analytic-cell sampling framework.
- [2] LAPACK, *DGESVD: singular value decomposition*, official routine documentation.
- [3] LAPACK Users' Guide, *Further Details: Error Bounds for the Singular Value Decomposition*.
- [4] NIST Digital Library of Mathematical Functions, Section 18.3 and Section 18.9: classical orthogonal polynomials and derivative identities.